

THE GPU SHOWDOWN

An evidence dossier for curious computer scientists

Plot Twist. Something is wrong with our GPU ranking. The newest card did not always win. One assignment was stopped before it began because the computer was not the GPU we thought it was. Your mission is to decide what the evidence actually supports.

The case file

We measured Aider/Ollama inference on a small fleet. The headline number is *observed whole-system inference latency*: wall-clock time shaped by the GPU, memory, model residency, CPU, operating system, backend, driver, load, and experimental conditions. It is not a silicon-only score.

The useful question is not “Which GPU wins?” It is: *What claim can this evidence carry, and what better test would make the claim stronger?*

Evidence Check. The automated percentage in the source reports is a concept coverage screen made from keyword/pattern checks. It is not a validated correctness grade.

Meet the machines

Machine	GPU	Family	CPU	VRAM (GB)	Warm s	Status
Machine A	RTX 5080	Blackwell	Intel Core Ultra 7 265 (20c)	16	2.48	safe
Machine B	RTX 2080 Super	Turing	AMD Ryzen 9 3950X (32c)	8	5.79	safe
Machine C	GTX 1660 Ti Max-Q	Turing	Intel Core i7-9750H (12c)	6	7.02	safe
Machine D	Tesla T4	Turing	Intel Xeon Gold 6252 (96c)	15	9.69	safe
Machine E	RTX 5060 Ti	Blackwell	Intel, Windows (20c)	8	10.80	caveated
Machine F	RTX 3060 Ti	Ampere	Intel, Windows (24c)	8	12.51	caveated
Machine G	GTX 1060 3GB	Pascal	AMD, Windows (24c)	3	14.73	caveated
Machine H	GTX 1080	Pascal	Intel Core i7-8700K (12c)	8	15.35	caveated

CPU pairing is included because it is one of the confounds this dossier keeps warning about: a GPU number alone does not fully explain observed whole-system latency. The values are curated from source experiment directories or immutable source-import packets, not copied from stale aggregate HTML. Machine E has 84 successful calls and 6 explicit timeouts; Machine F has 90 successful calls and no errors. Machine G has a residency limitation; Machine H’s context includes WSL2. All cross-machine comparisons remain caveated.

Confound Ahead. Machine B’s Turing RTX 2080 Super appears faster than the newer Blackwell RTX 5060 Ti and Ampere RTX 3060 Ti in these observed whole-system runs. That is a question generator, not proof that Turing is inherently faster than either newer architecture.

How the experiment worked

The original showdown used three prompt levels, five repetitions, six local models, wall-clock timing, raw outputs, hardware provenance, and an automated concept screen. A complete run targeted 90 calls. A timeout is a state, not a zero; a clean exit is not proof that an explanation is correct.

Supported observation	Claim we cannot make
Recorded systems had different observed latencies	A pure GPU-generation multiplier
Machine G's setup has a 3 GB residency limitation	An exact offload penalty without a controlled test
The gate found the wrong hardware	A numeric RTX 5060 benchmark

Trust Nothing. A direct hardware check stopped the intended “Machine E RTX 5060” assignment when the machine was actually an RTX 4070 with about 12 GB VRAM. Refusing to run the wrong experiment is a successful scientific outcome.

Computer Architecture lens

For architecture, treat the table as a whole-system observation. GPU generation is one factor among VRAM capacity, residency/offload, CPU/host behavior, backend, driver, and load. If a model spills from VRAM, the path can change; if software differs, the comparison is not controlled.

A useful calculation

For the observed warm values, a simple speedup is $S = T_{slow}/T_{fast}$. Use the table to calculate the Machine H/Machine B value yourself; the result supports only the statement that these recorded systems differed by that observed ratio in this setup.

The useful surprise

Machine B's Turing RTX 2080 Super is listed at 5.79 s, while the Machine E Blackwell RTX 5060 Ti is 10.80 s and the Machine F Ampere RTX 3060 Ti is 12.51 s. These are observed whole-system values: Machine E includes six explicit timeouts, and Machine F's separate 15.92 s all-six-model aggregate is not part of this three-common-model ranking. Newer architecture did not automatically win; the result invites controlled follow-up rather than a silicon-only conclusion.

Measure First. Claim Second.. Architecture explanations become stronger when the model, prompt, software stack, host, residency, repetitions, and load are controlled.

Your mission – Computer Architecture

Answer with numbers and evidence. Mark each claim as supported, caveated, or unsupported.

1. Calculate the observed speedup of Machine B over Machine H using the warm values. State why this is not a silicon-only speedup.
2. Rank Machine A, Machine B, Machine C, and Machine D from fastest to slowest. What observation is surprising, and what does it not prove?
3. Machine G has 3 GB of VRAM and a caveated row. Explain how model residency or CPU offload could affect latency without inventing an unseen measurement.
4. Identify two confounds in comparing Machine A and Machine B. Name one control for each.
5. Rewrite this unsupported claim: “The newer GPU is slower because Blackwell is worse than Turing.” Use observation, evidence, warrant, and caveat.
6. Explain why the wrong-hardware gate is a valid result. Why must it stay out of the numeric ranking?
7. Design a better controlled comparison: specify hardware, model, software/backend, prompts, repetitions, residency check, and failure policy.

Evidence appendix cue

Every published number traces to a source commit and experiment path in the accompanying provenance ledger. Keep that chain intact in your work.